

NORFOLK SOUTHERN RAILROAD BRIDGE N-647.74
Spanning Stony Creek at Milepost 647.74
Pride
Ross County
Ohio

HAER OH-141
HAER OH-141

PHOTOGRAPHS

WRITTEN HISTORICAL AND DESCRIPTIVE DATA

HISTORIC AMERICAN ENGINEERING RECORD

National Park Service
Interior Regions 3, 4, and 5
U.S. Department of the Interior
601 Riverfront Drive
Omaha, NE 68102

HISTORIC AMERICAN ENGINEERING RECORD
NORFOLK SOUTHERN RAILROAD BRIDGE N-647.74

HAER No. OH-141

Location:	Spanning Stony Creek at milepost 647.74 on the NS Columbus Division Pride (southeast of Chillicothe), Franklin Township, Ross County, Ohio The Norfolk Southern (NS) bridge over Stony Creek is located at latitude: 39.235822°, longitude: -82.893288°. The coordinates were taken from the northeast corner of the bridge and were obtained in March 2021 through Google Earth using the World Geodetic System 1984 (WGS84) geographic coordinate system (datum), which is accurate to approximately +/- 3 meters. U.S. Geological Survey (USGS) Quad: Waverly North, Ohio. The bridge location has no restriction on its release to the public.
Date(s) of Construction:	ca. 1888 (stone arch [center]); 1909 (stone and brick arch on west [inlet] side); and 1934 (reinforced concrete box extension on east [outlet] side). No records stating the exact date of construction for the original stone arch were available at the time this documentation was completed. The estimated date of construction of ca. 1888 aligns with a newspaper account describing an improvements campaign the Scioto Valley Railway Company (SV) undertook in the late 1880s that included the replacement of wooden bridges with new iron or stone structures along the company's alignment from Portsmouth to Columbus, Ohio.¹
Architect/Engineer/Builder:	The SV built the oldest of the bridge components, a 16' long stone arch, ca. 1888. In 1909, the Norfolk & Western Railway (N&W) added a second arch to the west (inlet) side of the 16' long arch, to allow for double tracking over the crossing. The N&W constructed the

¹ "Exchanges, Ross County Register," *Portsmouth Daily Times*, October 6, 1888.

third component of the current structure, a concrete box culvert east of the ca. 1888 arch, in 1934.

Original Owner/

Occupant and Use: SV, culvert carrying one main line track of railway.

Present Owner/

Occupant and Use: NS bridge carrying one main line track of railway.

Significance:

The Ohio History Connection, which serves as the State Historic Preservation Office, determined the NS bridge over Stony Creek (Bridge N-647.74) eligible for listing in the National Register of Historic Places in May 2020. The structure is significant at the local level under Criteria A and C. The structure is a representative example of a series of brick, stone, and concrete arches erected in the late nineteenth (ca. 1888) and early twentieth century (1909 and 1934). The SV likely built the oldest component of Bridge N-647.74, a stone arch culvert, while completing improvements to the line in the late 1880s. The improvements campaign included the replacement of wooden bridges with new iron or stone structures along the company's entire alignment from Portsmouth to Columbus, Ohio.² Subsequent additions—the west expansion of the inlet in 1909 with a second masonry arch and the east expansion of the outlet in 1934—occurred under the N&W's ownership of the bridge (July 1890–1981).

In the late nineteenth century, prior to the widespread use of reinforced concrete, railroad companies typically constructed arch bridges and culverts across the United States of stone, a material known for its durability. Often, stone arch bridges were built in areas where quality stone was readily available, such as in Waverly, Pike County, Ohio, approximately 9 miles south of the bridge site.³ No sources explicitly state that Waverly stone was used in the construction of Bridge N-647.74, but given the bridge's proximity to the source it is a possibility local stone was used. The concrete extensions of the bridge demonstrate typical modifications railroad companies made to stone structures as part of improvements and/or repairs across the nation in the early twentieth century. Of the stone arch bridges constructed in the United States during the late eighteenth and early nineteenth

² "Exchanges, Ross County Register," *Portsmouth Daily Times*, October 6, 1888.

³ Parsons Brinkerhoff. *A Context for Common Historic Bridge Types*, NCHRP Project 25-25, Task 15. (National Cooperative Highway Research Program, Transportation Research Council, National Research Council, October 2005), 3-50.

centuries, examples are considered highly significant if they retain their character-defining features, including arch rings with keystones, barrel, spandrel wall, parapet, headwalls, and abutments/wingwalls.⁴ The 1909 arch of Bridge N-647.74 retains the majority of those character-defining features, though the headwall was modified during the construction of the 1934 concrete extension. The original stone arch ring of the ca. 1888 arch is partially visible from the outlet (east end) of the structure.

Description:

The bridge at Milepost N-647.74 consists of three adjoining culverts built ca. 1888–1934. The bridge carries a single mainline track of the NS railroad across Stony Creek in Franklin Township, Ross County. The NS railroad runs northwest to southeast at the site of the bridge, which is situated in a narrow, forested corridor in a rural part of the county. Stony Creek flows west to east at the bridge site and then immediately into the Scioto River to the east. Three Locks Road (County Highway 205) makes a bend to the west approximately 0.10 mile southwest of Bridge N-647.74 and intersects with Stoney Creek Road (County Highway 199) approximately 0.20 mile southwest of the bridge.

The bridge measures approximately 64' in length from the face of the inlet headwall (west elevation) to the face of the current outlet headwall (east elevation); the wing walls on the east side of the structure extend an additional 26'-6" east of the outlet wall. The oldest of the bridge components is a 16' long stone arch built by the SV ca. 1888, which was designed to carry a single track. In 1909, the N&W added a second arch to the west (inlet) side of the 16' long arch, to allow for double tracking over the crossing. The third component of the current structure is a concrete box culvert east of the ca. 1888 arch, added in 1934. In 1934, when N&W completed the addition of the concrete box culvert, the double tracks were shifted to the east. The second track was removed sometime after 1960, returning the alignment to a single track over Bridge N-647.74. The single track lies on a graded slope approximately 4' to 6' above the headwall of the bridge.

History:

The SV formed February 23, 1875, with plans to build a main line between Columbus and Portsmouth, a shipping point on the Ohio River. Construction of the SV railway line began in Columbus in June 1875, and the line was extended south to Chillicothe by July 1876. The

⁴ Parsons Brinkerhoff, *A Context for Common Historic Bridge Types*, 350.

SV began operating between Chillicothe and Columbus in 1876 while the southern portion of the line was under construction.⁵

Chillicothe, the present-day county seat of Ross County, was platted in August 1796, the same year that Zane's Trace, the earliest road cut through the area, reached the town.⁶ The land comprising present-day Ross County was initially a part of Adams County, which was established in July 1797. In August 1798, Ross County was created.⁷ In May 1800, Congress named Chillicothe the seat of the territorial government. In early 1803, Ohio achieved statehood, and Chillicothe was named the state capital. The state capitol moved to Zanesville from 1810–1812, then returned to Chillicothe briefly before moving to Columbus permanently in 1816. Franklin Township was established in eastern Ross County on September 9, 1806, and its boundaries were adjusted to their present configuration in 1814.⁸

Following statehood, agriculture quickly became the center of the regional economy, for the soils of the Scioto River valley were ideal for the cultivation of staple crops such as corn and wheat. Flatboats first traveled down the Scioto River from Chillicothe to New Orleans, Louisiana, in 1803, transporting grain, flour, and livestock. Flatboats also traveled north to Circleville, Ohio, and Columbus carrying agricultural products. The first cattle drive took place in 1804 and traveled east from the Scioto Valley over the Allegheny Mountain Range to Baltimore, Maryland.⁹ Over the following decades, cattle of the Scioto Valley became highly sought after, and the local cattle industry thrived as a result.

Prior to the arrival of rail transit, canal construction was the first large-scale infrastructure project in Ohio. The state's General Assembly passed the first act related to canal construction in 1821. The Ohio and Erie Canal, completed 1825–1832, traveled through Ross County, and extended through the eastern portion Franklin Township before turning southwest to cross into Pike County. The State Dam across the Scioto

⁵ Striplin, E.F. Pat. *The Norfolk & Western: A History*. 1981. (Reprint, Norfolk: Norfolk Southern Corporation, 1997), 59–60.

⁶ Ross County Genealogical Society. *A Standard History of Ross County, Ohio, Volume I*. 1917. (Reprint, Baltimore: Gateway Press, Inc., 1987), 78.

⁷ Ross County Genealogical Society, *A Standard History of Ross County, Ohio, Volume I*, 52.

⁸ Ross County Genealogical Society. *State Centennial History of the County of Ross (Ohio), Volume II*. 1902. (Reprint, Baltimore: Gateway Press, 1981), 307.

⁹ Ross County Genealogical Society, *State Centennial History of the County of Ross (Ohio), Volume II*, 136.

River, designed to provide a water supply to the canal, was constructed south of Chillicothe, approximately 1.4 miles north of Stony Creek. Completed in 1832, the dam measured approximately 300' in length. Three locks were located adjacent to the dam, to feed water from the Scioto River to the canal.¹⁰ Later in the nineteenth century, Three Locks became the name of a nearby railroad stop, and is also the namesake of a road that travels along the river.

In the 1850s, railroads expanded across the state and quickly began posing competition to the existing canal system. After 1851, the state's canal system struggled to break even, under both state ownership and a brief period of operation by private lessees in the 1870s. During the same period, total rail mileage in the state grew from approximately 300 miles in 1850 to over 5,600 miles in 1880.¹¹

In 1877, the SV began to extend its line south from Chillicothe. The SV ran parallel to the Ohio and Erie Canal on the east side of Franklin Township. A railroad station known as both "State Dam" and "Three Locks" was located there. Other SV stations in the township in 1880 included Stony Creek, the county bridge, and Higby. In 1880, Higby was the only station in the township without a station house, though a post office was located in the community.¹² In 1881, the SV was completed to Coal Grove, Ohio, slightly east of Portsmouth, providing uninterrupted service between the Ohio River and Columbus.¹³ The SV primarily transported agricultural products of the Scioto Valley, and the extension to Coal Grove provided access to iron manufacturing and lumber traffic along the Ohio River.

The SV went into receivership in May 1885, and receiver Joseph Robinson continued operating and making improvements along the line until 1890. Improvements in the late 1880s included the replacement of wooden bridges with iron structures and construction of telegraph lines between twenty-three of the SV's twenty-four stations.¹⁴ Construction of the oldest stone arch culvert component of

¹⁰ Ross County Genealogical Society. *History of Ross and Highland Counties, Ohio with Illustrations and Biographical Sketches*. 1880. (Reprint, Interlaken: Heart of the Lakes Publishing, 1991), 281.

¹¹ Roseboom, Eugene H. and Francis P. Weisenburger. *A History of Ohio*. (Columbus: The Ohio Historical Society, 1976), 221.

¹² Ross County Genealogical Society, *History of Ross and Highland Counties, Ohio with Illustrations and Biographical Sketches*, 282.

¹³ Striplin, *The Norfolk & Western: A History*, 60.

¹⁴ Ibid.

Bridge N-647.74 likely occurred ca. 1888, under Robinson's receivership, a period during which stone culverts replaced many older wood structures along the existing line. It is possible that local stone was used to construct the ca. 1888 arch; the eastern portion of Ross County, like Pike County to the south, was rich in mineral deposits, including shale, limestone, and sandstone. In Franklin Township, one local history source reported "a great amount of valuable Waverly building stone exposed on the Scioto bank," that extended "for at least twenty miles along the Scioto, on both sides."¹⁵ Judiah E. Higby, a wagonmaker and flatboat builder and operator, invested in land in Franklin Township, and opened quarries along the Ohio and Erie Canal. However, no source consulted for this study explicitly states that local stone was used in the construction of the SV arch over Stony Creek.

In February 1890, the Scioto Valley and New England Railroad (SV&NE) acquired the SV, and, a few months later, in July 1890, the N&W acquired the SV&NE. The N&W named Robinson superintendent of the "Scioto Division," the name ascribed to the original SV route within the N&W system. The N&W's origins date to 1836, when the City Point Railroad was chartered in Virginia. Over the next forty-four years, the future N&W took shape through over 200 mergers. Those companies were consolidated under the Norfolk & Petersburg Railroad (N&P) by the 1860s. Following the Civil War, the N&P merged with two smaller railroads—the Southside and the Virginia & Tennessee—to form the Atlantic, Mississippi & Ohio Railroad (AM&O). By 1881, the AM&O had gone bankrupt, and E. W. Clark & Company, a bank in Philadelphia, Pennsylvania, purchased the N&P and renamed it the N&W. Frederick J. Kimball, a partner in E. W. Clark & Company, took leadership of the N&W and promptly merged it with the Shenandoah Valley Railroad, which ran between Hagerstown, Maryland, and Roanoke, Virginia.

Kimball, who had an interest in geology, was reportedly the catalyst behind the N&W opening the Pocahontas Coalfields in Virginia and West Virginia, and led the charge to build a rail connection with the SV&NE (the SV's "Ohio Extension") to provide a route to Columbus.¹⁶ The N&W completed its Ohio Extension in 1892, providing service to Columbus. In 1901, the N&W achieved service to

¹⁵ Ross County Genealogical Society. *State Centennial History of the County of Ross (Ohio), Volume II*, 131.

¹⁶ Middleton, William D., George M. Smerk, and Roberta L. Diehl, eds. *Encyclopedia of North American Railroads*. Indiana University Press: Bloomington, 2007), 750.

Cincinnati through the purchase of the Cincinnati, Portsmouth & Virginia Railroad.¹⁷ In 1902, the N&W purchased the Pocahontas Coal and Coke Company, and the number of mines along its route grew to over 140 only a few years later. Between 1905 and 1920, the amount of coal the N&W shipped increased from 8.18 to 26 million tons, and the amount of westbound shipments increased dramatically via the Ohio Extension.

In Franklin Township, Higby was the N&W's only stop in 1902. By 1906, a Stony Creek Station and Pride Post Office are noted on USGS topographic maps. By 1911, Pride was noted as a stop on the N&W at the approximate location of the Stony Creek Station listed in 1906, both situated south of the Locks stop (called Three Locks in other sources) and north of Higby, the southernmost N&W stop in Ross County.¹⁸ The N&W undertook a campaign of improvements in 1909, including double tracking through the county,¹⁹ and the name change from Stony Creek Station to Pride may have occurred during that time. The 1909 improvements included the expansion of the inlet of Bridge N-647.74 with a second masonry arch to accommodate a second main line track. A school was established in Pride in the early 1900s. Though diminished in use, a limited amount of commerce continued along the Ohio and Erie Canal until it was abandoned after suffering severe flood damage in 1907.²⁰

As railroad operations reached their zenith both locally and nationwide, roadway improvements, and the construction of highways to facilitate increasing numbers of automobiles, began spreading across the county. By 1912, five "Inter-County Highways" met at Chillicothe. The route of present-day U.S. Highway 23 was one such route, following the approximate north to south route of the Scioto Trail. U.S. Highway 23 serves as the primary north-south road through the county, running from Portsmouth north to Columbus and on to Toledo, Ohio. U.S. Highways 35 and 50 run northwest to southeast, and east to west, respectively, through Chillicothe. Beginning in the 1920s, highways began luring both passenger and freight traffic from

¹⁷ Ibid.

¹⁸ Rand McNally and Company. *Rand, McNally & Co.'s New Business Atlas Map of Ohio* [map]. 1:823,680. Chicago; United States Geological Survey. Waverly, Ohio Quadrangle Map (1:62,500).

¹⁹ Chillicothe Gazette. "N&W Will Double Track Higby's to Kingston." January 6, 1909; "Hard at Work." July 13, 1909; and "Railroad Notes." October 25, 1910.

²⁰ Ohio History Connection. "Chillicothe, Ohio." Ohio History Central. https://ohiohistorycentral.org/w/Chillicothe,_Ohio, accessed March 19, 2021.

railroads. The third and final expansion of Bridge N-647.74—the construction of the current concrete outlet with wingwalls—occurred in 1934.

While the N&W had historically traversed primarily rural areas of the county, by the mid twentieth century small communities that once were the site of railroad stations, such as Higby, Pride, and Three Locks, no longer served as stops. Though small pockets of residential development are located near Bridge N-647.74, vestiges of the canal and railroad activity of earlier times are limited. The Pride School closed in the late 1960s but remained extant until 2001 when a fire destroyed it.²¹ By the late 1990s, no towns or established businesses were present in Franklin Township.²²

As railroad mergers became increasingly common in the 1970s, the prospect of a merger between the N&W and Southern Railway arose. In 1982, the Interstate Commerce Commission approved the acquisition of the two lines by the newly formed Norfolk Southern Corporation. The NS and Southern continued operating under their historic names until 1990, when the N&W became a subsidiary of the Southern. In 1999, the system grew again after the NS and CSX purchased and split the Conrail system. As of 2005, the NS system totaled over 21,100 route miles of railway in twenty-two states; Washington, D.C.; and Ontario, Canada.²³ NS continues to operate the former SV alignment through Ross County.

Bridge N-647.74, in its current configuration, has been in continuous operation since 1934. In the late-2010s, structural inspectors identified stone failure within the arch culvert, including missing and damaged stones in the arch ring and interior, and across the inlet (west) elevation. Cracking and spalling are present on the concrete components of the structure. Project engineers have advised that most of the 100 year old stone likely would not survive the deconstruction and reconstruction necessary for rehabilitation. To stabilize the stone and brick arch culverts and repair the concrete box extension, the NS proposes installation of a corrugated steel arch liner with a reinforced concrete base that extends the width of the existing bridge, providing support for all three components. As part of the proposed project, concrete encasements would be constructed at the inlet and outlet

²¹ *Chillicothe Gazette*. “In Wake of Fire, Students Remember Former School.” January 30, 2001.

²² *Chillicothe Gazette*. “Franklin Township Not Big On Size But is Big on Family.” October 20, 1998.

²³ Yenne, Bill, *Atlas of North American Railroads*. (MBI Publishing Company, St. Paul, 2005), 129.

elevations, which measure 159 feet and 32.5 feet long, respectively. The stabilization efforts proposed would allow the historic components to stay in place, albeit totally enclosed and supported by the steel liner and concrete encasements.

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Historian(s):

Jessica M. Forbes, architectural historian (HDR Engineering, Inc.) completed the HAER report February–April 2021.

**Project
Information:**

NS sponsored the completion of this HAER documentation, which is required to be submitted to the Nebraska District of the U.S. Army Corps of Engineers to satisfy mitigation requirements for adverse effects associated with the repairs of the railroad bridge over Stony Creek, in accordance with Section 106 of the National Historic Preservation Act of 1966. Leslie Schwartz (Leslie Schwartz Photography) completed the large format photographs, and Jessica M. Forbes, architectural historian (HDR Engineering, Inc.), prepared the report. Leesa Gratreack, architectural historian (HDR Engineering, Inc.), and Nancy Jepsen, technical editor (HDR Engineering, Inc.), edited and assisted in formatting the document.

Supplemental Materials:

Location Map

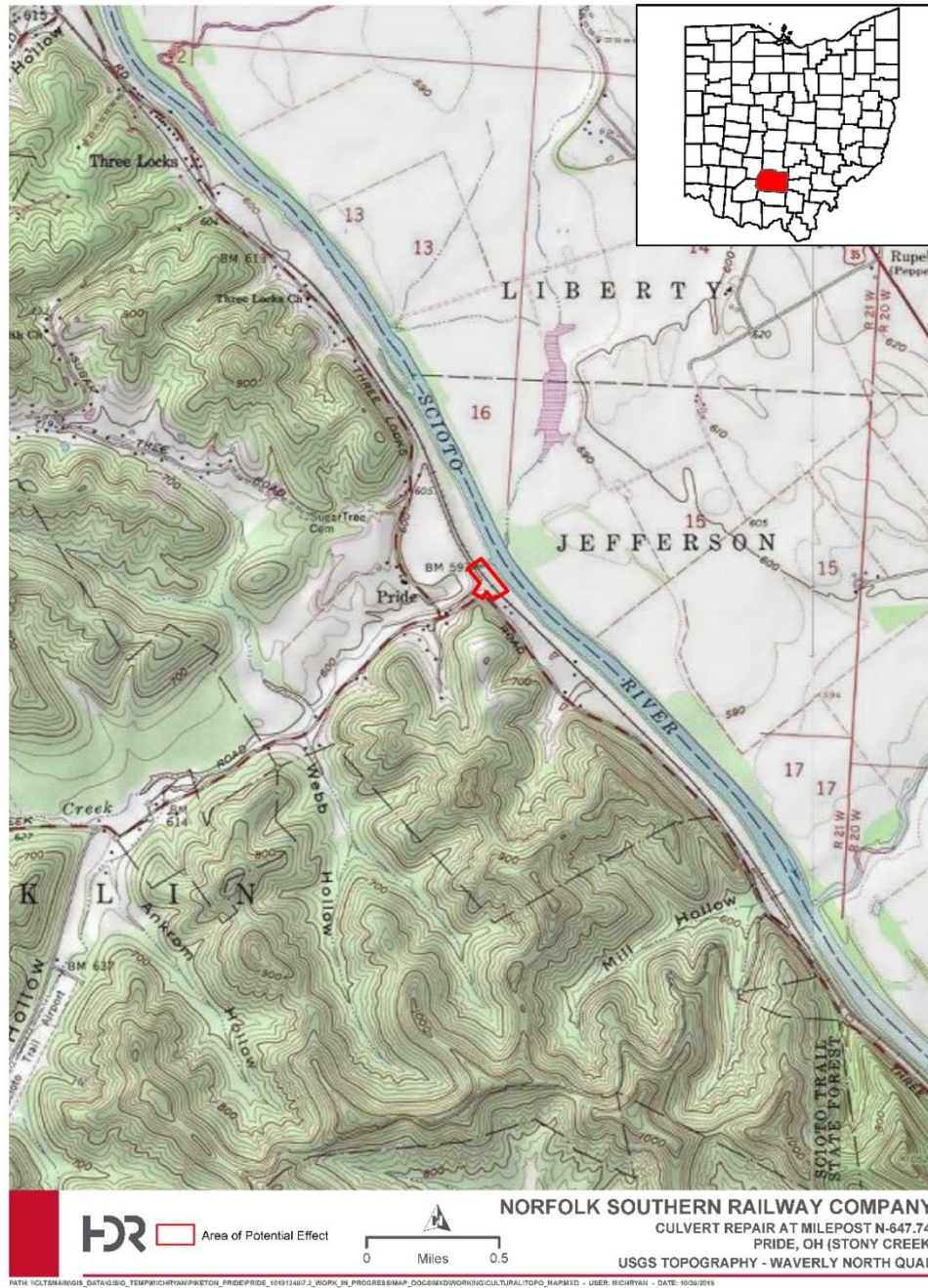


Figure 1. Bridge location.²⁴

²⁴ Figure prepared by HDR Engineering, Inc. using USGS 1992 Waverly North Quadrangle Map (1:24,000).

Color Photographs



Figure 2. East elevation (outlet/1934 extension), looking west (September 2019).²⁵

²⁵ Color photographs by HDR Engineering, Inc.



Figure 3. East elevation (outlet/1934 extension), looking west (September 2019).²⁶

²⁶ Color photographs by HDR Engineering, Inc.



Figure 4. Interior view(from background to foreground: 1909 arch, ca. 1888 arch, and 1934 extension), looking west (September 2019).²⁷

²⁷ Color photographs by HDR Engineering, Inc.



Figure 5. Arch detail (ca. 1888 arch ring partially visible), looking west (September 2019).²⁸

²⁸ Color photographs by HDR Engineering, Inc.



Figure 6. Arch detail (ca. 1888 arch ring partially visible), looking southwest (September 2019).²⁹

²⁹ Color photographs by HDR Engineering, Inc.



Figure 7. 1909 arch detail, looking southwest (September 2019).³⁰

³⁰ Color photographs by HDR Engineering, Inc.



Figure 8. 1909 arch detail, looking north (September 2019).³¹

³¹ Color photographs by HDR Engineering, Inc.



Figure 9. 1909 and ca. 1888 arch detail from west elevation (inlet), looking northeast (September 2019).³²

³² Color photographs by HDR Engineering, Inc.



Figure 10. 1909 and ca. 1888 arch detail from west elevation (inlet), looking southeast (September 2019).³³

³³ Color photographs by HDR Engineering, Inc.



Figure 11. Voids on west elevation (1909 arch and 1934 concrete header), looking east (September 2019).³⁴

³⁴ Color photographs by HDR Engineering, Inc.



Figure 12. Detail showing date stamps (1909 and 1934) on west elevation, looking east (September 2019).³⁵

³⁵ Color photographs by HDR Engineering, Inc.

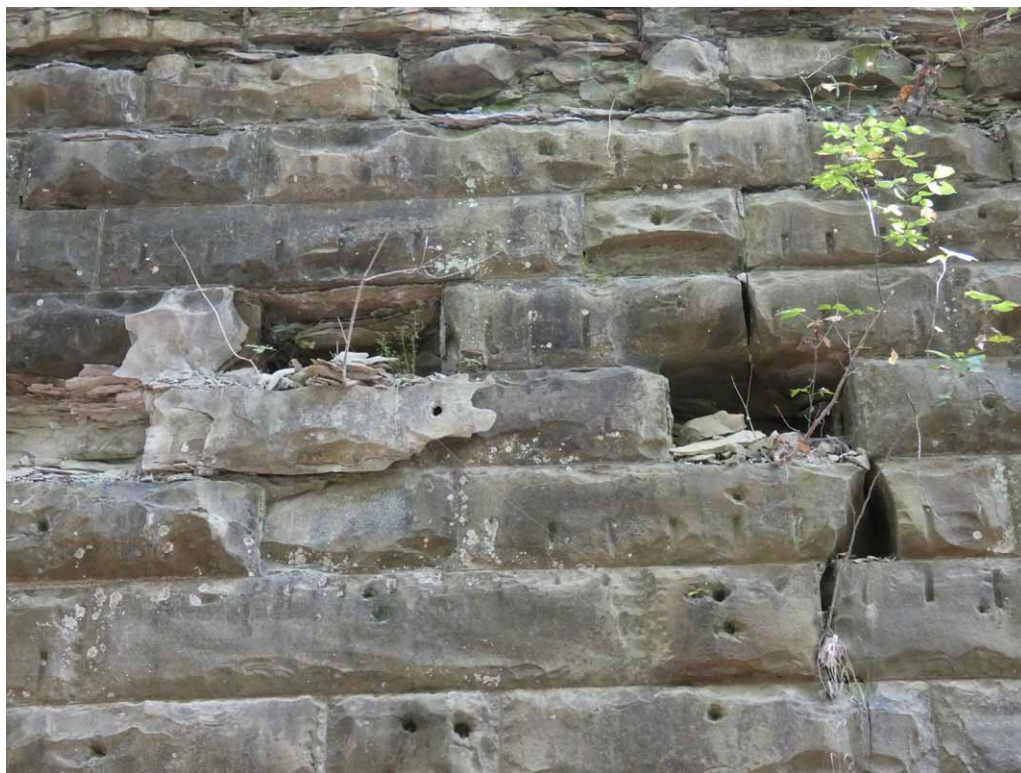


Figure 13. Voids on west elevation/1909 retaining wall, looking east (September 2019).³⁶

³⁶ Color photographs by HDR Engineering, Inc.

[illegible]

³⁷ “Plan of Masonry: Stony Creek Arch West of Higby, Ross County, O.” Norfolk & Western Railway. F.B. Z91 P. 4-5-4-6. July 19, 1907 (Source: Norfolk Southern Railway).

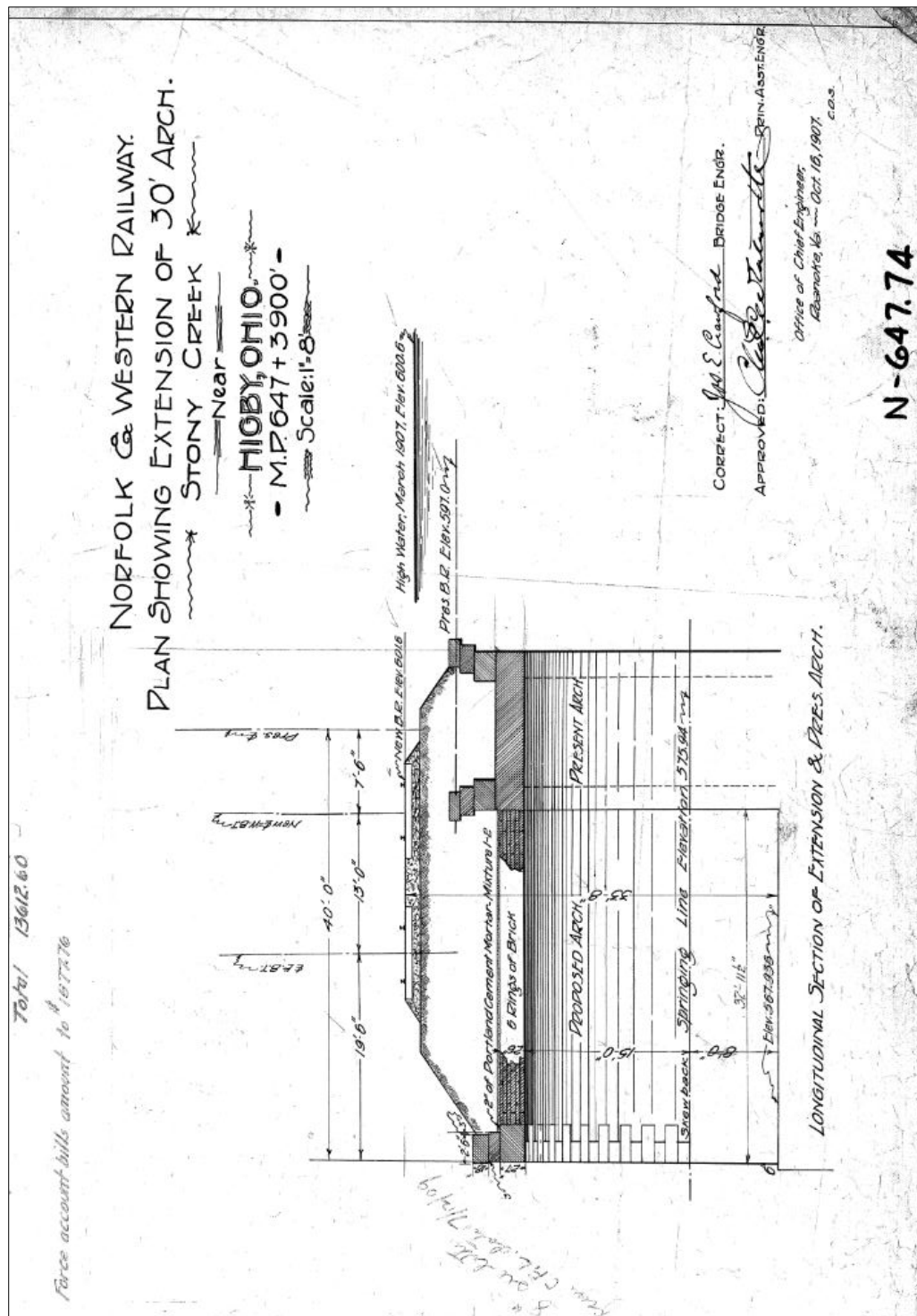
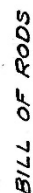


Figure 15. Detail of plan showing extension of Stony Creek arch, 1907.³⁸

³⁸ "Plan Showing Extension of 30' Arch, Stony Creek near Higby, Ohio." Norfolk & Western Railway. October 16, 1907 (Source: Norfolk Southern Railway).



³⁹ “Stony Creek Arch – 16 ft. Extension with 28’-6” x 19’0” Rein. Conc. Box Near Locks, O.” Norfolk & Western Railway Company. July 11, 1934 (Source: Norfolk Southern Railway).



⁴⁰ “Stony Creek Arch – 16 ft. Extension with 28’-6” x 19’0” Rein. Conc. Box Near Locks, O.” Norfolk & Western Railway Company. July 11, 1934 (Source: Norfolk Southern Railway).

Sketch Plan

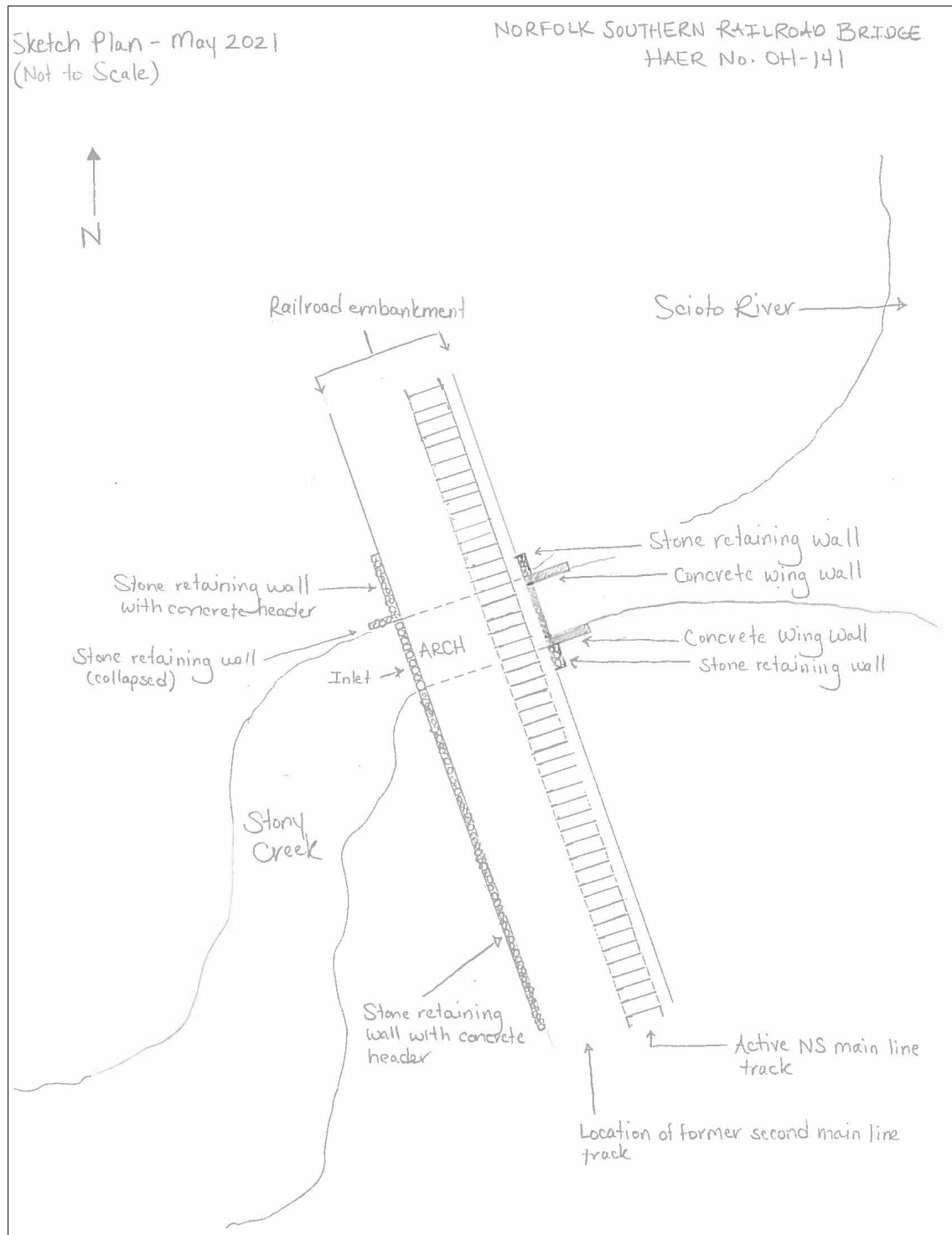


Figure 18. Sketch plan (May 2021). Prepared by HDR Engineering, Inc.